MIL–C–18345A
21 NOVEMBER 1962

SUPERSEDING
MIL–C–18345 (AER)
21 DECEMBER 1954

MILITARY SPECIFICATION

CORE MATERIAL, CELLULAR CELLULOSE ACETATE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the requirements for cellular cellulose acetate core material, having a density of six to seven pounds per cubic foot.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP–B–636 —Box, Fiberboard

MILITARY

MIL–P–116 —Preservation, Methods of
MIL–P–7936—Parts and Equipment, Aeronautical, Preparation for Delivery

STANDARDS

FEDERAL

FED–STD–406 —Plastics: Methods of Testing

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–129 —Marking for Shipment and Storage
MIL–STD–401 —Sandwich Construction and core Materials, General Test Methods

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois).

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sxiteenth Street, N.W., Washington 6, D. C.).

3. REQUIREMENTS

3.1 Material. The material used in the manufacture of core material shall be cellulose acetate. The cellular cellulose acetate core material shall conform to the applicable requirements as specified herein.

3.1.1 *Uniformity.* The core material shall be homogeneous throughout. The cells shall all be of approximately the same size. There shall be no visible large void openings or large accumulations of unexpanded resin or other occlusions.

3.2 Form and size. The cellular cellulose acetate core material shall be furnished in extruded sheets, blocks, or other forms, as

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specified by the procuring activity. The dimensions and dimensional tolerances shall be as specified in the contract or order and applicable drawings.

3.3 **Physical properties.** The core material shall conform to the physical properties specified in Table I, when tested as specified in 4.6.2.

TABLE I
Physical Properties

Property	Requirement
Density (lbs/cu. ft.)	6.5 ± 0.5
Tension, Flatwise, Ultimate Strength (psi)	250 (min)
Compression, Flatwise, Ultimate Strength (psi)	110 (Min)
Flexure, Flatwise, Ultimate Strength (psi)	160 (min)
Flexure, Flatwise, Modulus of Elasticity (psi)	5900 (min)
Two Plate Shear, Tensile, Ultimate Strength (psi)	100 (min)
Two Plate Shear, Tensile, Modulus of Rigidity (psi)	4000 (min)
Flammability (in./min.)	4.9 (max)
Temperature Resistance (°F)	(1)

(1) The core material shall withstand a temperature of 300° ± 5° F. without distortion, shrinkage or softening.

3.4 **Workmanship.** The cellular cellulose acetate core material shall conform to the dimensional requirements of the applicable drawings or as specified in the contract or order and shall be free from cracks, surface irregularities, injuries, or contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.**

4.2.1 *Acceptance inspection.* All inspection shall be to determine compliance with the requirements of this specification to serve as a basis for acceptance.

4.3 **Lot.** For purposes of inspection, a lot shall consist of all material manufactured at the same time, of one density and configuration, in the same production run, and offered for delivery at the same time.

4.4 **Sampling.**

4.4.1 *Sampling for visual and dimensional examination.* A random sample of core material shall be selected from each lot offered to the Government in accordance with Standard MIL–STD–105 at inspection level II. The acceptable quality level (AQL) shall be 1.5 per cent defective.

4.4.2 *Sampling for tests.* A random sample of core material shall be selected from each lot offered to the Government. Samples for the nondestructive test (see 4.6.2) shall be selected in accordance with Standard MIL–STD–105, inspection level II. The acceptable quality level (AQL) shall be 1.0 per cent defective. Samples for the destructive tests (see 4.6.2) shall be selected in accordance with Standard MIL–STD–105, inspection level I–4, acceptance number zero defects.

4.5 **Examination.**

4.5.1 *Visual and dimensional examination.* Each of the samples of core material selected in accordance with 4.4.1 shall be examined to verify conformance with 3.1, 3.2 and 3.4, and all other requirements which do not involve destructive tests.

4.6 **Testing.**

4.6.1 *Test conditions.* Unless otherwise specified, laboratory tests conditions shall be as specified in Standard MIL–STD–401.

4.6.2 *Test Methods.* Samples selected in accordance with 4.4.2 shall be subjected to the tests specified in Table II.

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4.7 Rejection.

4.7.1 *Visual and dimensional examination.* Any nonconforming piece in each sample when examined in accordance with 4.5.1 shall be rejected, and if the number of nonconforming pieces exceeds the acceptance number as specified for that sample, the entire lot shall be rejected.

4.7.2 *Testing.* When tested as specified in Table II, a lot shall be rejected if the number of nonconforming pieces exceeds the acceptance number of either the non-destructive test or the destructive tests.

4.8 Resubmittal of acceptance samples. When lots of core material are resubmitted for acceptance inspection in accordance with contractural directives, samples shall be selected in accordance with the provisions of Standard MIL–STD–105 employing tightened inspection.

TABLE II
Test Methods

Test	Method
Non-destructive	
Density	(2)
Destructive	
Tension, flatwise	(2)
Compression, flatwise	(2)
Flexure, flatwise (1)	(2)
Two plate shear, tensile	(2)
Flammability	FED–STD–406, Method 2021
Temperature Resistance	(3)

(1) This test shall be conducted on the core material alone. The formula for determining the ultimate strength shall be the same as listed in MIL–STD–401 as "flexural proportional limit" except that the load "P" shall be the ultimate load.

(2) Applicable paragraph in Standard MIL–STD–401.

(3) Place an accurately measured 4 inch cube in a controlled temperature cabinet maintained at 300° ± 5° F. for 24 hours. At end of this time the material shall be measured, and there shall be no indication of distortion, shrinkage or softening, when compared to a sample cut from the same source and not exposed to 300° ± 5° F.

4.9 Packaging, packing and marking. Preparation for delivery shall be examined for conformance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified in the contract or order, in accordance with the procedure outlined in MIL–P–7936. Level A material shall be packaged in accordance with MIL–P–116, Method III.

5.2 Packing. Packing shall be level A, B, or C as specified in the contract or order, in accordance with the procedures outlined in MIL–P–7936. Core material shall be packed in Type I or II or Class 1 or 2 fiberboard boxes conforming to PPP–B–636, within the size limitations of the boxes. If the size limitations of the fiberboard boxes are exceeded, appropriate exterior containers as specified in MIL–P–7936 shall be used.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Cellular cellulose acetate core material covered by this specification is intended for use in fabrication of structural sandwich parts.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Form and size of core material required (see 3.2).
(c) Quantity required.
(d) Applicable levels of packaging and packing required (see Section 5).

Notice. When Government drawings, specifications, or other data are used for any purposes other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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